

Reverse a String Using Recursion in Java

Code:

```
public class Main {  
    public static String reverse(String str) {  
        if (str.isEmpty()) {  
            return str;  
        }  
        return reverse(str.substring(1)) + str.charAt(0);  
    }  
    public static void main(String[] args) {  
        String str = "Hello";  
        System.out.println(reverse(str));  
    }  
}
```

Output: olleH

Time Complexity: $O(n^2)$ because substring() creates new strings in each recursive call.

Space Complexity: $O(n)$ due to the recursion call stack.